SDG BLOCKCHAIN ACCELERATOR

Mentor Feedback Report : Fuel Switch

Program: EMURGO : UNDP Blockchain Accelerator (Cohort 2)
Project: Fuel Switch
Focus Area: Renewable Energy Certification, Carbon Accounting, CBAM Readiness
Mentorship Focus: Technical Architecture, Regulatory Translation, System Design
Total Mentoring Sessions: 4
Mentor(s): Ahmed Hadded/ Cong Lee

1. Initial Context and Framing

Fuel Switch entered the accelerator with one of the clearest real-world problems in Cohort 2. The team was not exploring a speculative use of blockchain, but responding to a concrete regulatory shift: the introduction of the EU Carbon Border Adjustment Mechanism (CBAM) and its implications for electricity exports from Bosnia and Herzegovina.

From the outset, the team demonstrated a strong understanding of:

- How electricity is produced and exported in the Western Balkans
- Why the shared grid makes renewable and non-renewable electricity indistinguishable
- Where existing certification and reporting processes fall short under CBAM

What was less clear at the start was how to translate these regulatory and accounting requirements into a system that could be enforced technically, rather than asserted through reports or declarations. The mentoring work therefore focused on turning regulatory intent into concrete system constraints.

2. Mentoring Sessions

Session 1 : Translating Regulation into System Rules

The first session focused on aligning the regulatory problem with technical design choices. Rather than starting from blockchain primitives, the discussion began with CBAM itself: what exactly needs to be demonstrated, what level of granularity is expected, and what an auditor would realistically need to see.

Early design ideas leaned toward high-level certificates representing large volumes of renewable energy. While conceptually simple, this approach raised concerns around auditability and over-issuance. A single large certificate is easy to issue but difficult to scrutinize.

The key outcome of this session was a shift in mindset. Certificates were reframed not as symbolic representations of “green energy,” but as accounting instruments that must withstand scrutiny. This led to the decision to represent renewable energy claims as discrete, capped units that could be independently examined, retired, and traced.

The team agreed that regulatory credibility would come from constraints and limits, not from dashboards or narratives.

Session 2 : Certificate Lifecycle and Smart Contract Integrity

The second session moved into the details of smart contract design. The focus here was the lifecycle of a Renewable Energy Certificate: how it is created, who is allowed to create it, how it is used, and how it is definitively retired.

One recurring theme was the risk of double counting. If a certificate can be reused, partially consumed, or ambiguously retired, it fails its core purpose. The discussion therefore centered on making misuse technically impossible rather than procedurally discouraged.

This session resulted in several important design decisions. Minting authority was tightly restricted, with explicit on-chain checks rather than backend assumptions. Each certificate was capped at a fixed maximum energy value to avoid vague aggregation. Retirement was modeled as a destructive action (burn) with explicit safeguards to prevent any increase in supply during the burn process.

The mentor feedback emphasized that clarity and rigidity at this stage would pay dividends later, especially when interacting with regulators or auditors unfamiliar with blockchain systems.

Session 3 : Off-Chain Orchestration and Failure Modes

In the third session, attention shifted away from the smart contract itself and toward the surrounding infrastructure. The goal was to understand how the system behaves when things go wrong: retries, partial failures, batch processing, and network issues.

The team's backend design was generally sound, but some assumptions about request handling and retries posed risks. In particular, there was a possibility that repeated API calls or backend restarts could trigger multiple mint attempts for the same underlying data.

Rather than relying on "best effort" handling, the mentoring discussion pushed toward explicit idempotency and deterministic behavior. The backend was positioned as an orchestration layer, not a source of truth. Any operation that mattered had to be safe to repeat without unintended side effects.

This session also touched on operational realities: how batch minting might scale, where transaction costs could become an issue, and how to keep the system understandable for non-technical stakeholders reviewing outputs.

Session 4 : Cost, Performance, and Pilot Readiness

The final session focused on readiness rather than correctness. By this point, the core architecture was sound, but questions remained about performance and long-term viability.

The NFT-based approach chosen for the PoC was acknowledged as clear and auditable, but potentially expensive at scale. Rather than attempting premature optimization, the mentoring discussion emphasized sequencing: start with a model that is easy to reason about, validate it in pilots, and only then consider optimizations such as fungible representations for bulk issuance.

Testnet results were reviewed, including successful minting and retirement flows and the behavior of unauthorized actions. The overall assessment was that the system behaves predictably and defensively, which is the correct posture for a compliance-oriented solution.

The session concluded with a shared understanding that the current design is appropriate for controlled pilots and stakeholder demonstrations, with a clearly documented path for future evolution.

3. Overall Mentor Assessment

Fuel Switch is a strong example of a team that understands its regulatory problem space and is willing to let that understanding shape technical decisions. Over the course of the mentorship, the project moved from a conceptual certification idea to a defensible system design with clear enforcement points and auditability.

The team showed particular strength in:

- Grounding technical decisions in regulatory reality
- Accepting constraints where they improve trust and clarity
- Engaging constructively with feedback and adjusting designs accordingly

Remaining challenges are not architectural but strategic: managing costs as issuance scales, communicating system guarantees to non-technical stakeholders, and navigating the boundary between technical certification and formal regulatory acceptance.

Overall, Fuel Switch is well positioned for pilot engagement and further collaboration with institutional and regulatory stakeholders, provided the current discipline around clarity and constraint is maintained as the system evolves.